Enterprise Network Design

Cisco Packet Tracer

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by

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Project Requirements & Scope

Hardware Requirements

The simulated topology includes the following minimum equipment inventory:

- End Devices: 20 Workstations, 3 Laptops, and 2 Employee Mobile Devices.
- Networking Hardware: 3 Switches, 2 Routers, and 1 Firewall.
- Infrastructure: Wireless Access Points for mobile connectivity.
- Servers: 3 Servers covering various roles (Local Server, WAN Server, Web Server).

Logical & Configuration Requirements

The network architecture is designed to meet these specific logical constraints:

- Subnetting: Creation of at least two Sub-LAN networks, each utilizing different Class C IP addressing ranges and unique subnet masks.
- DMZ Implementation: A dedicated Demilitarized Zone containing a Local Server and at least one PC, separated from the main internal LANs.
- WAN Connectivity: A simulated Wide Area Network containing a server configured with a Class A IP address range.
- Network Services:
  - Full connectivity (Ping) between all workstations and servers.
  - A functional Web Server hosting a website.
  - A DNS Server correctly resolving the website URL.

- Security: Placement of a firewall in the optimal location to filter traffic between the WAN and the internal network.

Technical Solution & Network Design

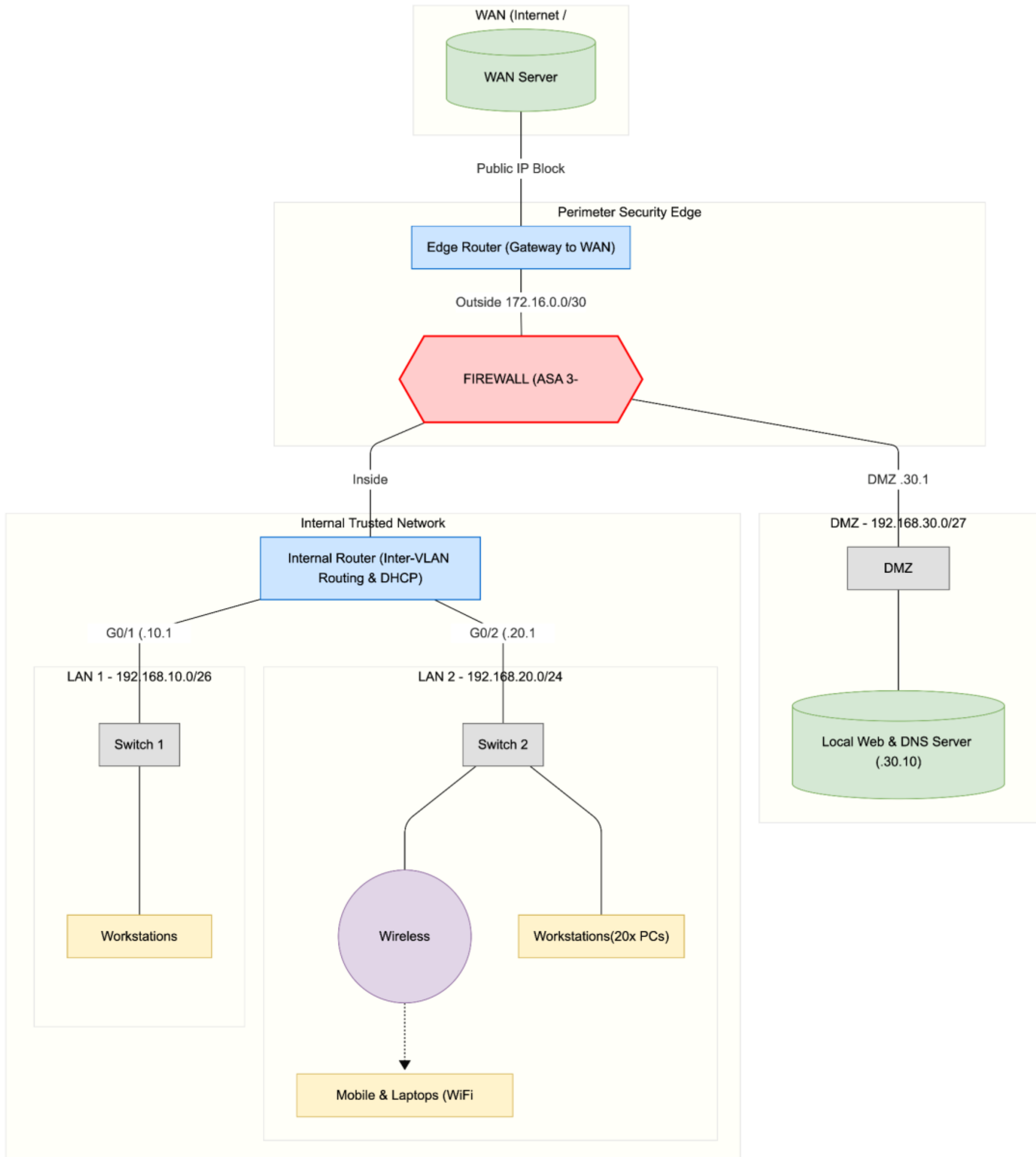
Design Philosophy

The technical solution proposed in this design follows the Hierarchical Network Design Model. This approach breaks the network into modular layers, which simplifies management, improves fault isolation, and increases scalability.

- Access Layer: Managed by the switches (Cisco 2960), this layer connects end-user devices (PCs, Laptops) to the network. It controls user access and provides initial segmentation via VLANs or physical separation.
- Core/Distribution Layer: Managed by the Routers (Cisco 2911), this layer handles routing between the different subnets (LAN 1, LAN 2, and DMZ) and provides high-speed switching.
- Edge/Perimeter Layer: Managed by the Firewall and Edge Router, this layer acts as the gateway to the WAN (Internet), strictly controlling inbound and outbound traffic.

Block Diagram

The following block diagram illustrates the high-level logical topology of my solution. It demonstrates the traffic flow from the external WAN, through the firewall, and into the segmented internal zones (DMZ and LANs).



Hardware Selection Rationale

To ensure the network meets feature requirements, I selected the following Cisco devices for the simulation:

- Router (Cisco 2911): Selected for its integrated services and support for Gigabit Ethernet interfaces. It allows for Inter-VLAN routing or physical interface routing, which is essential for connecting the Class C LANs with the Class A WAN.
- Switch (Cisco 2960): A Layer 2 fixed-configuration switch chosen for its reliability and ease of configuration. It provides sufficient port density to support the 20 workstations and access points required.
- Firewall (ASA 5506-X): Chosen to sit at the network perimeter. In a real-world scenario, this device performs Stateful Packet Inspection (SPI). For this simulation, it is placed logically between the ISP Router and the Internal Router to demonstrate the correct security architecture.
- Server (Generic PT Server): Capable of running multiple simultaneous services (HTTP, DNS, DHCP) within the simulation environment, reducing the need for excessive hardware while demonstrating full functionality.

Network Segmentation Strategy

The solution avoids a “flat” network topology, where all devices share a single broadcast domain. Instead, the network is segmented into three primary zones:

- Zone A (Internal LANs): Contains the 20 workstations and employee devices. This is the “Trusted” zone. It uses Class C private addresses (e.g., 192.168.10.0/24) to ensure sufficient address space for the required hosts.

- Zone B (DMZ): Contains the public-facing Web Server. This is a “Semi-Trusted” zone. If an external attacker compromises the web server, the strict routing rules prevent them from easily accessing Zone A.
- Zone C (WAN): The “Untrusted” external internet, simulated by the Class A network (10.0.0.0/8).

This segmentation is achieved using distinct Router interfaces and subnets, ensuring that traffic between zones must pass through the router and firewall, allowing for monitoring and control.

IP Addressing Scheme

Subnetting Strategy

Wide Area Network (WAN) - Public Simulation

- Network: 8.8.8.8
- Subnet Mask: 255.255.255.0/24

The Edge Router connects to an external server representing a public internet service. I utilized the 8.8.8.0/24 range to simulate a public Class A block, clearly distinguishing external traffic from internal private traffic.

Point-to-Point Backbone Links

- Internal Router to Firewall: 10.0.0.0/30
- Firewall to Edge Router: 172.16.0.0/30

High-efficiency masks were chosen for router-to-firewall connections. A /30 subnet provides exactly 2 usable host IP addresses. This eliminates wasted IP addresses and enhances security by preventing any other devices from attaching to these critical backbone links.

Internal LANs - Class C with VLSM

To adhere to the requirement of distinct Class C subnets while optimizing for network size, I applied VLSM:

- LAN 1 (Workstations): 192.168.10.0/24 – Capacity: 254 hosts. This is the primary user network.
- LAN 2 (Operations): 192.167.20.0/26 – Capacity: 62 hosts. A tighter subnet mask is used here to limit the broadcast domain for this specific department.
- DMZ (Servers): 192.168.30.0/27 – Capacity: 30 hosts. The DMZ is strictly limited to 30 hosts to minimize the attack surface.

Addressing Table

The topology centers around the Firewall, which acts as the gateway for the DMZ and the bridge between the internal and edge routers.

Device / Segment	Interface	IP Address	Subnet Mask	CIDR	Description
Edge Router	g0/2	8.8.8.1	255.255.255.0	/24	Gateway to Internet

WAN Server	g0/1	8.8.8.8	255.255.255.0	/24	Public External Server
Edge Router	g0/1	172.16.0.1	255.255.255.252	/30	Link to Firewall
Firewall	g1/2 (Outside)	172.16.0.2	255.255.255.252	/30	Link to Edge Router
Firewall	g1/1 (Inside)	10.0.0.1	255.255.255.252	/30	Link to Internal Router
Internal Router	g0/3	10.0.0.2	255.255.255.252	/30	Uplink to Firewall
Internal Router	g0/1	192.168.10.1	255.255.255.0	/24	Gateway for LAN 1
Internal Router	g0/2	192.168.20.1	255.255.255.192	/26	Gateway for LAN 2
Firewall	g1/3 (DMZ)	192.168.30.1	255.255.255.224	/27	Gateway for DMZ
DMZ Web Server	g0/1	192.168.30.10	255.255.255.224	/27	Public Web/DNS Server

Topology Note

In this design, the Firewall serves as the central traffic enforcement point.

- DMZ Termination: The DMZ network (192.168.30.0) is directly connected to the Firewall. This ensures that any traffic attempting to leave the DMZ towards the LANs is immediately inspected.

- Double Routing: Traffic from the internal LANs travels to the Internal Router, then to the Firewall, then to the Edge Router.

Configuration Details

To satisfy the requirement that any Work Station or Server can ping any other equipment, specific configurations were applied to the routers, switches, and server applications. The configuration ensures connectivity across the multi-layered topology (LAN - Internal Router - Firewall - Edge Router - WAN).

Routing Strategy

Given the specific topology where the Firewall sits between two routers, a Static Routing approach was implemented to ensure predictable traffic flow. This avoids the overhead of dynamic protocols like OSPF for this specific size of network.

Internal Router Configuration

The Internal Router acts as the gateway for LAN 1 and LAN 2. A Gateway of Last Resort was configured to send all non-local traffic to the Firewall. If a PC in LAN 1 tries to access the Internet or the DMZ, the router forwards the packet to the Firewall's inside interface.

Edge Router Configuration

The Edge Router sits between the ISP/WAN and the Firewall. It needs to know how to reach the internal networks. Routes were added pointing to the Firewall's outside interface. This summary route ensures that traffic returning from the WAN destined for any internal subnet (LAN 1, LAN 2, or DMZ) is correctly handed off to the Firewall.

DHCP

To manage the 20 workstations and mobile devices efficiently, DHCP Services were configured on the Internal Router. This eliminates the need to manually configure IP addresses on every employee device.

- Pool "LAN1":
  - Network: 192.168.10.0
  - Default Gateway: 192.168.10.1
  - DNS Server: 192.168.30.10
- Pool "LAN2"**:**
  - Network: 192.168.20.0
  - Default Gateway: 192.168.20.1
  - DNS Server: 192.168.30.10

By distributing the DMZ Server's IP 192.168.30.10 as the DNS server via DHCP, all internal clients can automatically resolve the website URL (mysite.de).

Application Services

The Server located in the DMZ (192.168.30.10) was configured to handle both HTTP and DNS responsibilities, consolidating resources.

Web Server (HTTP)

HTTP and HTTPS services were enabled. The default index.html file was edited to display "Hello World" to serve as proof of configuration during testing.

DNS Server

DNS service was enabled. A A-Record was inserted with the name of "mysite.de" at address 192.168.30.10. When a user types the URL into the web browser, the DNS server resolves it to the local IP address.

Wireless Connectivity

To support the "Employees Mobile devices" and "Laptops," a WAP was connected to Switch 2 (LAN 2). Mobile devices connected successfully and received IP addresses from the LAN 2 DHCP pool (192.168.20.x), providing full integration with the wired network.

Security Implementation

Security is a primary pillar of this network design. Rather than relying solely on router ACLs, the architecture incorporates a dedicated Firewall and a DMZ. This strictly controls traffic flow based on security levels.

DMZ Architecture

The network implements a specialized subnet designated as the DMZ. This zone contains the Public Web Server and DNS Server. By placing the public servers in a separate VLAN/Subnet, I ensure that the internal LANs are not directly exposed to the internet. If an external attacker manages to compromise the Web Server via the public internet, they are "trapped" in the DMZ. They cannot easily pivot to the internal network because the Firewall sits between the DMZ and the Internal Router, blocking unauthorized cross-zone traffic.

Firewall Placement

The firewall is deployed in a Three-Legged Topology. This is the optimal location for an enterprise edge firewall as it acts as the central traffic police for three distinct zones.

The three interfaces are:

1. The "Outside" Interface (Connected to Edge Router):
   - Security Level: 0 (Untrusted)
   - Function: Receives traffic from the WAN/Internet. The firewall is configured to block all unsolicited traffic entering this interface, except for specific requests destined for the Web Server (HTTP/Port 80).
2. The "Inside" Interface (Connected to Internal Router):
   - Security Level: 100 (Trusted)
   - Function: connects to the sensitive internal LANs. Traffic originating from here is generally allowed to go out (to WAN or DMZ), but traffic entering here is strictly scrutinized.
3. The "DMZ" Interface (Connected to DMZ Switch):
   - Security Level: 50 (Semi-Trusted)
   - Function: Holds the public servers. It allows traffic in from the WAN (for web access) but strictly denies traffic from the DMZ to the Inside interface.

Security Policy Logic

Although the assignment constraints limited the depth of firewall configuration, I had to configure the Firewall to integrate with the network. The design supports the following security policies which are standard for this topology:

- Rule 1 (WAN to DMZ): Permit HTTP/HTTPS traffic. (Allows the public to see the website).
- Rule 2 (WAN to LAN): Deny all traffic. (Prevents external attacks on employee PCs).
- Rule 3 (LAN to WAN): Permit all traffic. (Allows employees to access the internet).
- Rule 4 (LAN to DMZ): Permit DNS/HTTP. (Allows employees to view the company site and update the server).
- Rule 5 (DMZ to LAN): Deny all traffic. (Prevents a compromised server from attacking the internal network).

Testing & Validation Results

To verify the integrity and functionality of the network design, a comprehensive series of test cases was executed within Cisco. These tests cover connectivity with "ping", routing logic, and application services. I also used the Simulation environment primarily to debug and test things.

The following matrix summarizes the connectivity tests performed. A "Pass" indicates that ICMP Echo Requests (Ping) were successfully sent and received.

Source Device	**Destination Device**	**IP Address**	**Protocol**	**Result**	**Verification**

PC (LAN 1)	PC (LAN 2)	192.168.20.4	ICMP	Pass	Internal Routing OK
PC (LAN 1)	DMZ Web Server	192.168.30.10	ICMP	Pass	DMZ Access OK
Laptop (LAN 2)	WAN Server	8.8.8.8	ICMP	Pass	NAT/Routing to Internet OK
Mobile Phone	Default Gateway	192.168.10.1	ICMP	Pass	Wireless - Wired Link OK
Edge Router	Internal Router	10.0.0.2	ICMP	Pass	Backbone Link OK

Detailed Test Scenarios

Test Case 1: Inter-Departmental Routing (LAN to LAN)

- Objective: Confirm that the Internal Router is correctly routing traffic between the HR (LAN 1) and Operations (LAN 2) subnets.

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  Procedure: Open Command Prompt on PC-HR (192.168.10.10) and ping PC (192.168.20.10).
- Result: The ping was successful.
- Evidence:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=42ms TTL=127
Reply from 192.168.20.10: bytes=32 time=26ms TTL=127
Reply from 192.168.20.10: bytes=32 time=32ms TTL=127
Reply from 192.168.20.10: bytes=32 time=40ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 26ms, Maximum = 42ms, Average = 35ms
```

Test Case 2: WAN Connectivity (Internet Access)

- Objective: Verify that internal employees can reach the external internet (WAN Server).
- Procedure: From a Laptop in LAN 2, ping the WAN Server at 8.8.8.8.

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- Analysis: The packet successfully traversed the Switch - Internal Router - Firewall - Edge Router - WAN Server, and returned. This confirms that the default static routes and firewall “permit” rules are functional.

Evidence:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:

Request timed out.
Reply from 8.8.8.8: bytes=32 time=11ms TTL=125
Reply from 8.8.8.8: bytes=32 time<1ms TTL=125
Reply from 8.8.8.8: bytes=32 time<1ms TTL=125

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 11ms, Average = 3ms
```

Test Case 3: Wireless Integration

- Objective: Ensure Mobile Devices connected via the WAP can access the wired network.
- Procedure: From “Smartphone0”, ping the DMZ Server (192.168.30.10).
- Result: Successful reply. This proves the WAP is correctly bridging the wireless frames to the wired Ethernet switch.

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Evidence:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.30.10: bytes=32 time=30ms TTL=126
Reply from 192.168.30.10: bytes=32 time=25ms TTL=126

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 25ms, Maximum = 30ms, Average = 27ms
```

Test Case 4: Web Server & DNS Resolution

- Objective: Verify that the DNS Server correctly resolves the URL mysite.de to the IP address 192.168.30.10 and that the Web Server delivers the HTML page.
- Procedure: Open the Web Browser application on any internal PC and type "mysite.de" into the URL bar and press "Go".
- Result: The browser displayed the custom web page "Hello World".

- Evidence:


Web Browser
X
<
>
URL
http://hellofromnoel.de
Go
Stop
Hello from Noël

References

Cisco Networking Academy. (2025). *CCNA: Introduction to Networks & Switching, Routing, and Wireless Essentials*. [Online]. Available at: https://www.netacad.com